CHAPTER 2

Limitations on follow-up

S14, S16, S19, S20

Before you start · Inequalities and geometric sequences

Without a calculator, in six minutes. Answer briefly. Mobilize calculation on fractions and fractions then choose the targeted pick-up indicated under the questions.

1. Framework $\frac{1}{n+1}$ for $n \geq 1$.
2. Compare 0.7^{n+1} and 0.7^n for $n \geq 0$.
3. Give the sign of $\sqrt{n^2+1} - n$.
4. Streamlining $\sqrt{x} - 1$.
5. Is a increasing sequence necessarily enhanced?
6. Give an example of a non-constant bounded sequence.

Responses

$0 < \frac{1}{n+1} \leq \frac{1}{2}$; $0{,}7^{n+1} < 0{,}7^n$; positive ; multiply by $\sqrt{x}+1$; no; for example $(-1)^n$.

Targeted review after the diagnostic check

Repeat only the missed questions. For each question: **1.** Take reverses of two positive numbers; R16 recalls the effect of the sign. **2.** Write the following term as 0.7 times the Previous; R02. **3.** Compare positive squares before roots. **4.** Multiply by the conjugate and develop its product. **5.** Test the sequence of term n; distinguish variation and bound. **6.** Test powers of -1 in six rows; R02. R-questions serve as support for review with the Professor, even if their collective ritual is planned later.

Objectives

Distinguish convergence and divergence; calculate a limit; use comparison, gendarmes and monotonous convergence.

Course notes

Definitions

$u_n \to \ell \in \mathbb{R}$ means that any open interval containing ℓ contains all terms from a It is said that (u_n) converges to ℓ. $u_n \to +\infty$ means that for all real A, there is one row N such that $n \geq N \Rightarrow u_n \geq A$. $u_n \to -\infty$ means that for all real A, there is N such that $n \geq N \Rightarrow u_n \leq A$.

A sequence may have no limit, even infinity: $(-1)^n$ oscillation between -1 and 1. Any further action converging is bounded, but the converse is false.

Operations and comparisons

For two converging sequences of limits a, b, the sum tends to $a+b$, the product ab and quotient $\frac{a}{b}$ if $b \neq 0$. Rules with infinity are used by controlling signs.
$+\infty - \infty$, $0 \times \infty$, $\frac{\infty}{\infty}$, $\frac{0}{0}$ do not allow for direct conclusion.

Example. $\frac{3n^2-n+1}{2n^2+4} = \frac{3-\frac{1}{n}+\frac{1}{n^2}}{2+\frac{4}{n^2}} \to \frac{3}{2}$. For $\sqrt{n^2+n}-n$, multiply by the conjugate quantity given $\frac{1}{\sqrt{1+\frac{1}{n}}+1} \to \frac{1}{2}$.

Theorems

If $u_n \leq v_n$ from a certain rank and $u_n \to +\infty$, then $v_n \to +\infty$. If $a_n \leq u_n \leq b_n$ from a certain rank and $a_n, b_n \to \ell$, then $u_n \to \ell$ (gendarmes) Any increasing and increased succession converges. Any decreasing and minor sequence converges. This theorem is allowed.

Demonstration by comparison. For every A, the terms u exceeding A beyond a row N_1. Inequality $v_n \geq u_n$ equals beyond a row N_2. For $n \geq \max(N_1, N_2)$, $v_n \geq A$.

Demonstration: increasing not increased. For every A, the non-majority provides N such that $u_N > A$. By growth, $n \geq N \Rightarrow u_n \geq u_N > A$, where $u_n \to +\infty$. Non-majority alone is not enough.

Geometrical sequences

$$q^n \longrightarrow \begin{cases} 0 & \text{if } -1 < q < 1, \\ 1 & \text{if } q = 1, \\ +\infty & \text{if } q > 1. \end{cases}$$

If $q \leq -1$, (q^n) no limit: alternation for $q = -1$; paired and odd sub-sequences of signs and unbounded modules for $q < -1$.

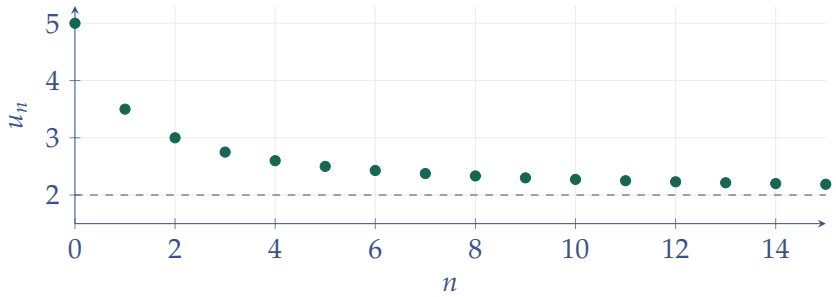
Bernoulli and geometric limit. Fixes $a \geq 0$. Demontrons $P(n)$: « $(1+a)^n \geq 1+na$ In row 0, the two members are worth 1. $n \geq 0$ and suppose $P(n)$ Like one. $+a > 0$,

$$(1+a)^{n+1} \geq (1+na)(1+a) = 1+(n+1)a+na^2 \geq 1+(n+1)a.$$

It's $P(n+1)$ since $na^2 \geq 0$. By induction, inequality is true for all $n \geq 0$. If $q > 1$, take $a = q-1 > 0$ and compare to $1+na$. If $0 < q < 1$, apply this result to $\frac{1}{q}$Then take the reverses. For $-1 < q < 0$, $|q^n| = |q|^n \to 0$ and the gendarmes conclude. $q = 0$ deals from row 1.

Recursively defined sequence

For $u_{n+1} = f(u_n)$, establish a bound and monotonicity to prove convergence. f is continues in the limit ℓ, the passage to the limit gives $\ell = f(\ell)$ (continuity discussed in Chapter 4). Solving this equation never proves, on its own, that the sequence converges.



Points close to 2 without reaching: $u_n = 2 + \frac{3}{n+1}$.

Give meaning to the limits

2.1 One band and all subsequent ranks

First use: S14.

Definition · Converging to ℓ

Next (u_n) converges to ℓ if, for any accuracy $\varepsilon > 0$, there is a row N such that $n \geq N$ implies $|u_n - \ell| < \varepsilon$. The rank depends on the accuracy, but once chosen, it is suitable for all the following terms.

Solved example · Calculate a valid rank

For $u_n = 2 + \frac{3}{n+1}$, $|u_n - 2| = \frac{3}{n+1}$. Precision 0 ,1 requires $n + 1 > 30$, so $n \geq 30$. Rank 29 is not suitable: the difference is exactly 0 ,1. For any precision, any N such that $N + 1 > 3/\varepsilon$ It's okay.

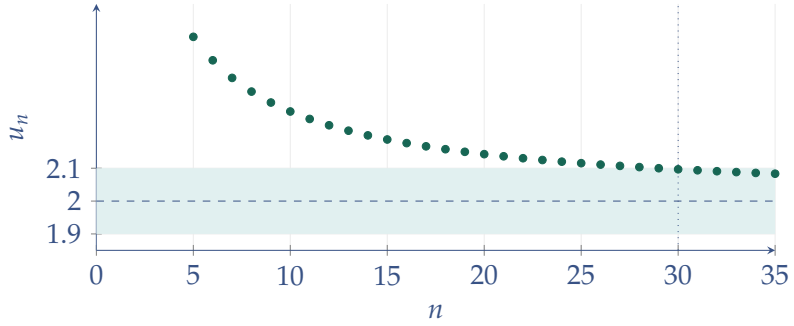


Figure 2.1. Band $]1{,}9;2{,}1[$: all terms of rank $n \geq 30$ stay there.

A1 · Immediate application For $v_n = 5 - \frac{2}{n+1}$, determine the first ranking from which $4{,}95 < v_n < 5.05$. Check the previous row and explain why all of the following agree.

A2 · Counterexample "There is a term in each band around 0, so the sequence converges to 0. » Refuse with a sequence whose even terms are null and the odd terms are worth 1.

2.2 Transform before concluding

First use: S16 and S19.

Solved Example · A Form Does Not Set a Limit

The three quotients n/n, n/n^2 and n^2/n all have a shape ∞/∞. Their limits are however 1, 0 and $+\infty$. For $\frac{3n^2-n+1}{2n^2+4}$ divide numerator and denominator by n^2 gives the limit $\frac{3}{2}$. The transformation is valid for $n \geq 1$, which is enough for a limit.

A3 · Choose before calculating Determining the limits of $\frac{2n+3}{n^2+1}$, $\sqrt{n^2+n}-n$ and $\frac{(-1)^n}{n+1}$. Name, for Each, transformation or decisive theorem.

A first step: In the second expression, multiply by the conjugate quantity. In the third, frame the absolute value.

2.3 Converging and then identifying

First use: S20; continuity identification after S32.

Ownership · Monotonous Convergence · Accepted

A increasing and increased sequence converges towards a real. The theorem does not say that it reaches its or that its limit is any of its majors.

Example Solved · A complete string

Let $u_0 = 1$ and $u_{n+1} = \sqrt{2+u_n}$. The interval $[1;2]$ is stable because $1 \leq u_n \leq 2$ results $\sqrt{3} \leq u_{n+1} \leq 2$. Over this interval, $2 + x - x^2 = (2-x)(x+1) \geq 0$, therefore $u_{n+1} \geq u_n$ After comparison positive squares. The sequence converges to $\ell \in [1;2]$. After continuity, $\ell = \sqrt{2+\ell}$ causes $(\ell-2)(\ell+1) = 0$; only $\ell = 2$ belongs to $[1;2]$.

A4 · Repairing a Common difference / ratio Next $a_0 = 1$, $a_{n+1} = 2a_n$ checks a relationship whose only point 0. A student concludes $a_n \to 0$. Show his error and determine the actual behavior.

A5 · Independent evidence Let $b_0 = 0$, $b_{n+1} = \frac{1}{3}b_n + 2$. Prove $0 \leq b_n < 3$, then growth. deduce convergence and determine the limit by the explicit formula or, after S32, by continuity.

2.4 Don't forget the negative reasons

First use: S19 and S21.

Property · Behaviour of q^n

If $|q| < 1$, $q^n \to 0$. If $q > 1$, $q^n \to +\infty$. For $q = 1$, the sequence is constant. $q \leq -1$, it does not no limit, even infinite: the signs alternate, and the two sequences of even and odd ranks have incompatible behaviors.

Demonstration. For $q > 1$, write $q = 1 + a$ with $a > 0$. Recurrence gives $(1+a)^n \geq 1 + na$, which tends to $+\infty$; the comparison concludes. $< q < 1$, apply this result to 1 $/q$ and take The other way around. $-1 < q < 0$, $|q^n| = |q|^n \to 0$ results $q^n \to 0$ per frame. $q = 0$ is immediate from row 1. □

Example Solved · Common difference / ratio and Behaviour

For $q = -2$, the same terms 4 ^{k} tend to $+\infty$ and oddness -2×4^{k} To $-\infty$: None common limit. $q = -1/2$, the signs also alternate, but the module 2 $^{-n}$ tend towards 0; the sequence converges.

A6 · Compare properties For $(-1)^n$, $-\frac{1}{n+1}$ and n, indicate: bounded ? monotonous ? convergent ? Give justifications, not just a table of answers.

Wait a minute at the courthouse. Explain why "increasing" does not mean "positive" or "trend towards $+\infty$ ».

Review postponed · S33 then S66. Re-identification by continuity; for a complete sequence, reconstruct the same bound chain, monotonicity, convergence, identification.

Research activity

A limit is not a last term · 30 min.

What's next? (u_n) defined on $\mathbb{N}$ by $u_n = 2 + \frac{3}{n+1}$.

1. Calculate u_0, u_9 and u_{99}. Is there an integer n such that $u_n = 2$? Justify.

2. Determine the smallest integer N such as, for the integer $n \geq N$, $|u_n - 2| < 0{,}01$.

3. Either $\varepsilon > 0$. Give, depending on ε, one integer N such that $|u_n - 2| < \varepsilon$ for every $n \geq N$.

4. We define on $\mathbb{N}$ the rest (v_n) by $v_n = 2 + \frac{(-1)^n}{n+1}$. Demonstrate that it converges to 2 and that What does this example show?

Progressive exercises

E1 · Direct application. We define on $\mathbb{N}$ the follow-up (u_n) and (v_n) by $u_n = \frac{4n-1}{n+2}$ and $v_n = n^2 - 7n$. Determine their limit by justifying the transformations used.

E2 · Application and justification. We define $u_n = \sqrt{n+1} - \sqrt{n}$ for every $n \in \mathbb{N}$. Using the the combined quantity, determine the limit of (u_n).

E3 · Method to be organised. We define $u_n = \frac{\cos n}{n}$ for the integer $n \geq 1$. Framework u_n, and then deduct from it limit of (u_n).

E4 · Choice of strategy. Next (u_n) is increasing and verifies $u_n < 5$ for all $n \in \mathbb{N}$. Justify that Is it necessarily equal to 5? Justify.

E5 · Synthesis or demonstration. We define $u_n = 3 - 2(-0{,}6)^n$ for every $n \in \mathbb{N}$. Determine the limit of (u_n). Is this sequence monotonous?

Work in the manual or annal

Barbazo, p. 62, Exercises 17 and 24. In S16: Exercise 17, to lift an indeterminate form by dividing by a power of n. At home after S20: Exercise 24, to write an application of the theorem Gendarmes.

Duty at home

DM2 · Approaching a root · 50 min.

It is considered $u_0 = 0$ and, for all $n \in \mathbb{N}$, $u_{n+1} = \sqrt{2 + u_n}$. Questions 1, 2 and 4 are to be dealt with after S20; question 3 will be resumed after the course on continuity in S31.

1. Demonstrate by induction that all terms are defined and that $0 \leq u_n \leq 2$ for all $n \in \mathbb{N}$.

2. Demonstrate by induction that $u_{n+1} \geq u_n$ for every $n \in \mathbb{N}$. Deduce that (u_n) converge.

3. Using the continuity of the function $x \mapsto \sqrt{2 + x}$, determine the limit of (u_n).

4. Demonstrate $0 \leq 2 - u_{n+1} \leq \frac{2 - u_n}{2}$ for every $n \in \mathbb{N}$. Deduct an increase of $2 - u_n$ in function of n, then one integer N such as $0 \leq 2 - u_n \leq 10^{-3}$ for every $n \geq N$.

MCQ A · definitions and operations

After S16. MCQ B after S20; fixed point after S31.

For each question, only one proposal is accurate.
consult the correction.

1. $u_n = 3 + \frac{2}{n+1}$. The limit is:

A. 0 B. 2

C. 3 D. $+\infty$

2. $\frac{2n^2+1}{5n^2+n}$ tend towards:

A. $\frac{2}{5}$ B. $\frac{5}{2}$

C. 0 D. $+\infty$

3. $u_n \to 2$ and $v_n \to 0$. What can be said without any other hypothesis?

A. $u_n / v_n \to +\infty$ B. $u_n v_n \to 0$

C. $u_n / v_n \to 0$ D. $u_n + v_n \to 0$

4. For $u_n = 2 + \frac{3}{n+1}$, the first ranking from which $|u_n - 2| < 0.1$ is:

A. 28 B. 29

C. 31 D. 30

5. Next $(-1)^n$ is:

A. converging to 0 B. Limited and diverging

C. Increasing D. diverging $+\infty$

QCM B · choose a theorem

For each question, only one proposal is accurate.
consult the correction.

6. We know. u_n increasing and $u_n \leq 5$. Necessarily:

A. $u_n \to 5$ B. $u_n \geq 0$

C. $u_n \to \ell \leq 5$ D. u_n constant

7. $u_n = \frac{\sin n}{n}$ for $n \geq 1$. The direct tool is:

A. Monotonous convergence

B. A fixed point equation

C. The law of large numbers

D. gendarmes

8. A increasing sequence not increased:

A. tend towards $+\infty$

B. may tend to 0

C. is always geometric

D. oscillate between two values

9. $\sqrt{n^2+n} - n$ tend towards:

A. $+\infty$

B. 1

C. $\frac{1}{2}$

D. 0

10. If $0 \leq u_n \leq (\frac{3}{4})^n$ at any level:

A. u_n is geometric

B. $u_n \to 0$

C. u_n is decreasing

D. $u_n \to \frac{3}{4}$

True or false · expected justification. For each statement, indicate whether it is true or false, and then write evidence or a counter-example.

1. Any converging sequence is monotonous.

2. Any increasing and bounded sequence converges.

3. If $u_n \to \ell$ and $u_n < 5$, then $\ell < 5$.

Corrected MCQs

1. **C.** The quotient tends to 0, so the sum tends to 3. Neither the numerator nor the first term Give the limit.

2. **A.** Divide numerator and denominator by n^2. The dominant terms are of the same degree.

3. **B.** The product tends to 2 $\times 0 = 0$. The quotient may be undefined, or change a sign; the sum tend to 2.

4. **D.** $3/(n+1) < 1/10$ is equivalent to $n > 29$. In row 29, the difference is exactly 0.1.

5. **B.** The same terms are worth 1, the odd -1 ; they cannot stay in the same small band The sequence is bounded.

6. **C.** Monotonous convergence. Constant sequences equal to -1 refute positivity and limit 5; $5 - 1/(n+1)$ Refuses constancy.

7. **D.** $-1/n \leq u_n \leq 1/n$, and both bounds tend towards 0; no monotonicity is required.

8. **A.** For every threshold, a term exceeds it, as do all later terms since the sequence is increasing. This is the definition limit $+\infty$.

9. **C.** By conjugation, the expression is $1 / (\sqrt{1+1/n}+1)$ for $n > 0$, where $1/2$. Subtraction Infinites did not allow to conclude.

10. **B.** The upper bound tends towards 0. A sequence between the bounds can go up on time, e.g. by cancelling the same terms.

True/false. False: $(-1)^n/(n+1) \to 0$ is not monotonous. True by monotonous convergence. False: $5 - 1/(n+1) \to 5$; only wide inequality passes to the limit.

Corrected applications and support

Application 1. The difference at 5 is $2/(n+1)$; we need $n+1>40$, so $n \geq 40$. In row 39, $v_{39} = 4{,}95$ is Excluded. Variations decrease, resulting in validity for all subsequent ranks.

Application 2. The following has terms equal to 0 in any band, but for $\varepsilon = 1/2$, any row is followed by an odd row whose term is 1. N ensures the universal condition.

Implementation 3. The first quotient tends to 0 by division by n^2. The second expression is $\frac{n}{\sqrt{n^2+n}+n} = \frac{1}{\sqrt{1+1/n}+1} \to 1/2$. The third has an absolute value equal to 1 $/(n+1) \to 0$, so tend towards 0.

Implementation 4. $a_n = 2^n \to +\infty$. Equation $\ell = 2\ell$ is a necessary condition if a finite limit exists; it does not prove this existence.

Application 5. Initially $b_0 = 0$. If $0 \leq b_n < 3$, then $2 \leq b_{n+1} < 3$, therefore the invariant is transmitted. $b_{n+1} - b_n = \frac{2}{3}(3 - b_n) > 0$. Monotonous convergence; the formula $b_n = 3 - 3(1/3)^n$ gives the limit 3. Out of continuity, $\ell = \ell/3 + 2$ also gives 3 after establishing convergence.

Implementation 6. $(-1)^n$ is bounded between -1 and 1, not monotonous and not converging because its sub-sequences pairs and odds have different limits. $-1/(n+1)$ is increasing, bounded between -1 and 0, and converging towards 0. n is increasing, not increased and diverges to $+\infty$.

Support points

Color the words that remain outside the band, then translate "remain" with the quantifiers.
Prior to S32, give continuity as an acceptable result in an identification exercise
or use an explicit formula. Proof of monotonous convergence itself remains accepted;
its implementation must be reusable.

Conduct and correction

Benchmarks for the teacher

The DS2 precedes S19 and S20: assess definitions and operations already encountered, without require monotonous convergence, Bernoulli or gendarmes. Draw horizontal bands around the limit and search for a row from which all the terms remain. convergence before solving the fixed point equation. Show $(-1)^n$ for dissociating "bounded" and "converging".

Corrected activity

1. $u_0 = 5$, $u_9 = 2.3$ and $u_{99} = 2.03$. Since $\frac{3}{n+1} > 0$ at any level, $u_n > 2$: no term equals 2.

2. $|u_n - 2| < 0{,}01 \iff \frac{3}{n+1} < 0{,}01 \iff n+1 > 300 \iff n \geq 300$. The first appropriate ranking is so $N = 300$. At range 299, the gap is exactly 0.01 and strict inequality is not satisfied.

3. For $\varepsilon > 0$, let's take it $N = \lceil \frac{3}{\varepsilon} \rceil$ the smallest integer greater than or equal to $\frac{3}{\varepsilon}$. If $n \geq N$, then $n+1 > \frac{3}{\varepsilon}$, so $|u_n - 2| < \varepsilon$. This choice is appropriate, even when it is not minimal.

4. $|v_n - 2| = \frac{1}{n+1} \to 0$, therefore $v_n \to 2$. But $v_0 = 3$, $v_1 = \frac{3}{2}$ and $v_2 = \frac{7}{3}$: one sees a decrease and then a This converging sequence is not monotonous.

E1. For $n \geq 1$, $\frac{4n-1}{n+2} = \frac{4-\frac{1}{n}}{1+\frac{2}{n}} \to 4$, because the denominator tends to $1 \neq 0$. Similarly $n^2 - 7n = n^2(1 - \frac{7}{n})$; the first factor tends to $+\infty$ and the second about $1 > 0$, so the product tends to $+\infty$.

E2. The conjugate gives $\sqrt{n+1} - \sqrt{n} = \frac{(n+1)-n}{\sqrt{n+1}+\sqrt{n}} = \frac{1}{\sqrt{n+1}+\sqrt{n}}$. The denominator tends to $+\infty$, therefore the difference tends to 0.

E3. For $n \geq 1$, $-1 \leq \cos n \leq 1$ and $n > 0$, therefore $-\frac{1}{n} \leq u_n \leq \frac{1}{n}$. The two bounds tend towards 0; the theorem of the gendarmes gives $u_n \to 0$.

E4. The continuation is increasing and increased by 5, so converge. By passing to the limit in $u_n \leq 5$, its limit check $\ell \leq 5$. It is not necessarily equal to 5: $u_n = 2 - \frac{1}{n+1}$ is increasing, increased by 5 and converging to 2.

E5. Like $|-0{,}6| < 1$, $(-0{,}6)^n \to 0$, therefore $u_n \to 3$. However $u_0 = 1$, $u_1 = 4{,}2$ and $u_2 = 2{,}28$: Then go up and down and is not monotonous.

Corrected from DM2

1. Note: $P(n)$: « u_n is defined and $0 \leq u_n \leq 2$". This is true in row 0 since $u_0 = 0$. Fixes $n \geq 0$ and Suppose $P(n)$ We've got two $\leq 2 + u_n \leq 4$, so $u_{n+1} = \sqrt{2+u_n}$ is defined and $\sqrt{2} \leq u_{n+1} \leq 2$. So $P(n+1)$ is true. By induction, all the terms belong to $[0,2]$.

2. Poisons $R(n)$: « $u_{n+1} \geq u_n$ For $n = 0$, $u_1 = \sqrt{2} \geq 0 = u_0$. Fixing $n \geq 0$ and suppose $R(n)$ The growth of $x \mapsto \sqrt{2+x}$ on $[0,2]$ gives

$$u_{n+2} = \sqrt{2+u_{n+1}} \geq \sqrt{2+u_n} = u_{n+1}.$$

So $R(n+1)$ holds. By induction, (u_n) is increasing. It is increased by 2, so converge to a real $\ell \in [0,2]$.

3. The function $x \mapsto \sqrt{2+x}$ is continuous in ℓ. By passing to the limit, $\ell = \sqrt{2+\ell}$. Like $\ell \geq 0$, this is equivalent to $\ell^2 - \ell - 2 = 0$, either $(\ell - 2)(\ell + 1) = 0$. Root -1 is excluded; therefore $\ell = 2$.

4. Using the conjugate and $u_n \leq 2$,

$$0 \leq 2 - u_{n+1} = \frac{2 - u_n}{2 + \sqrt{2 + u_n}} \leq \frac{2 - u_n}{2}.$$

Note: $T(n)$: « $0 \leq 2 - u_n \leq \frac{2}{2^n}$ In row 0, 2 $- u_0 = 2$. If $T(n)$ is true for a fixed integer $n \geq 0$, then

$$0 \leq 2 - u_{n+1} \leq \frac{1}{2}\frac{2}{2^n} = \frac{2}{2^{n+1}}.$$

Initialisation and inheritance prove $T(n)$ Any way. $\frac{2}{2^{11}} = \frac{1}{1024} < 10^{-3}$, $n = 11$ is enough.
It is a rank guaranteed by an increase, without a statement of minimality.
Before question 3: Number 2 is already involved in the framework.

Corrected work in the manual or an anal

Fiscal year 17. The quotients are first of indeterminate form. $n \geq 1$,

$$u_n = \frac{2n - 4}{n^2 + 1} = \frac{2\left(1 - \frac{2}{n}\right)}{n\left(1 + \frac{1}{n^2}\right)} \longrightarrow 0.$$

Similarly,

$$v_n = \frac{2 - 5n}{4n + 7} = \frac{\frac{2}{n} - 5}{4 + \frac{7}{n}} \longrightarrow -\frac{5}{4}, \qquad w_n = n\frac{-1 - \frac{10}{n^2} + \frac{4}{n^3}}{2 + \frac{3}{n} + \frac{1}{n^2}} \longrightarrow -\infty.$$

The last quotient tends to $-\frac{1}{2}$, and factor n To $+\infty$.

Fiscal year 24. For every $n \geq 1$, $-1 \leq (-1)^n \leq 1$. Since $n^2 > 0$, we get $-\frac{1}{n^2} \leq u_n \leq \frac{1}{n^2}$. Both limits tend towards 0; the theorem of gendarmes gives $u_n \to 0$.